ALGAMASK

Complete Project Guide for a Non-Technical Video Editor

A simple-English explanation of the idea, user journey, dataset, RAG system, six-agent workflow, experimental formulation engine, app screens, limitations, and video storyboard.

VIDEO PRODUCTION BRIEF

Prepared for Mazharul Islam and the AlgaMask team

Project status: functional local development system; scientific production approval is still required

01. The project in one sentence

Start the video with this idea before showing technical details.

AlgaMask is a research-support app that turns a person's health and food-preference information into a traceable, evidence-linked, experimental algae-based food formulation and a readable report.

The easiest way to explain it

The user answers a short health and preference questionnaire. The system searches its scientific knowledge base, passes the case through six specialist software agents, creates an experimental algae-food proposal, checks it, and produces a report that explains what was used and why.

Important language for the video

- Say **research-support system**, **experimental formulation**, and **evidence-linked suggestion**.
- Do not call it a doctor, a diagnostic system, a medical treatment, or a finished commercial food product.
- The current build is suitable for local development and demonstration. Human scientific review is still necessary before real production use.

The core story

Stage	What happens	Simple visual
1. Ask	Collect the user's goals, diet, allergies, and constraints.	Questionnaire screen
2. Search	Find the most relevant scientific records through RAG.	Documents moving into a search funnel
3. Think	Six specialist agents inspect the same case from different angles.	Six connected cards
4. Formulate	Build an experimental algae-based recipe with a baseline and controls.	Ingredient sliders or recipe card
5. Explain	Generate a traceable report with evidence and warnings.	Report with citations

02. The problem AlgaMask is trying to solve

Personalized food formulation is difficult because many things must be considered at the same time: a person's needs, ingredient science, allergies, taste, nutritional goals, formulation stability, and the quality of the evidence. A normal chatbot may sound confident without showing where its claims came from.

AlgaMask addresses that gap in four ways

Problem	AlgaMask response
Scientific information is scattered	A structured dataset stores sources and traceable scientific records.
Generic answers ignore the person	The profile captures goals, restrictions, allergies, and preferences.
One AI response is hard to inspect	Six agents create a visible step-by-step reasoning trace.
Claims can be unsupported	RAG retrieves evidence before the system writes explanations.
A recipe alone is not enough	The report includes formulation details, warnings, citations, and limitations.

What makes the project special

It combines personalized input, a scientific knowledge base, retrieval-augmented generation, multiple specialist agents, and an experimental formulation workflow in one traceable product.

Who could benefit

- Researchers exploring algae-based functional food concepts.
- Product-development teams comparing experimental formulations.
- Students and reviewers who need to see how a recommendation was produced.
- Eventually, approved professionals working with validated data and governed production rules.

03. The four major parts of the system

A. Scientific dataset

A structured collection of source documents and extracted records. Each record keeps enough information to trace a claim back to its source.

B. Local RAG

The retrieval system searches the dataset for the best evidence for the current user and returns only relevant passages to the reasoning workflow.

C. Six-agent backend

Six named software agents handle profile understanding, retrieval, chemistry, formulation, criticism, and impact reporting.

D. Flutter application

The user-facing app collects information, starts a run, displays progress and agent traces, and presents the final report and run history.

How data moves through the product

Input	Processing	Output
Profile and constraints	Validation and safety checks	Normalized user case
Scientific records	Embedding and similarity search	Evidence passages
Profile plus evidence	Six-agent orchestration	Traceable decisions
Agent decisions	Formulation and report assembly	Experimental recipe and report
Completed run	SQLite persistence and backup	History, retry, restore

Suggested animation

Show information flowing left to right: User -> Flutter App -> API -> RAG Knowledge Base -> Six Agents -> Formulation -> Report. Keep each label on screen long enough for a non-technical viewer to read it.

04. The scientific dataset

This is the foundation of RAG and traceability.

The project currently contains a development dataset organized into **21 tables with 286 records**. The records represent scientific knowledge needed by the system, such as ingredient properties, nutritional values, formulation constraints, and source-linked evidence.

What a traceable record means

- The record has a clear identity and structured fields.
- It records where the information came from, rather than storing an unsupported statement.
- The system can retrieve the record during a run and connect it to the generated explanation.
- A reviewer can inspect the evidence path from the final report back to the source record.

What is complete and what is pending

Area	Current state	Meaning
Schema	Implemented	The information has a consistent structure.
Development records	Available	The local app can run and demonstrate retrieval.
Traceability	Implemented	Records and report evidence can be connected.
Scientific approval	Pending	Experts must verify records for production use.
Production governance	Pending	Versioning, approvals, and controlled releases must be formalized.

Critical message for the video

The dataset is strong enough to power a functional development demonstration, but it must not be presented as fully approved medical or production-grade knowledge.

05. RAG explained in simple English

RAG means **Retrieval-Augmented Generation**. It helps the system look up relevant evidence before it creates an answer.

Think of it as an open-book process

Step	What it does	Plain-English comparison
1. Prepare	Convert scientific records into searchable numerical representations called embeddings.	Build a smart index for a library.
2. Ask	Turn the current user's needs into a search query.	Ask the librarian a focused question.
3. Retrieve	Find the most relevant records by meaning, not only exact words.	The librarian selects the best pages.
4. Ground	Give those records to the agents as evidence.	The team works with the selected pages open.
5. Cite	Attach evidence details to explanations and the report.	The reader can check the sources.

Why RAG matters here

- It reduces unsupported claims by grounding explanations in stored evidence.
- It lets the team improve the knowledge base without rewriting the whole application.
- It produces a visible retrieval trace for testing and review.
- It separates scientific knowledge from the wording of the final report.

Voice-over line

Instead of letting AI answer from memory, AlgaMask first searches its scientific library. It then gives the best-matching evidence to the specialist agents, so the final explanation is easier to inspect and verify.

06. The six specialist agents

1. Profile Agent

Reads the user's goals, diet, allergies, preferences, and constraints. It turns the questionnaire into a clean case summary.

2. Retrieval Agent

Builds the evidence query and asks RAG for the scientific records that best match the case.

3. Chemistry Agent

Examines ingredient properties, compatibility, composition, and formulation-related scientific considerations.

4. Recipe Agent

Builds an experimental formulation using the profile, evidence, and available formulation rules.

5. Critic Agent

Challenges the proposal. It checks conflicts, missing evidence, allergy issues, unrealistic values, and weak reasoning.

6. Impact Agent

Explains the likely purpose, trade-offs, cautions, evidence, and overall result in the final report.

Why use six agents instead of one chatbot?

The separation makes the process easier to understand, test, and improve. Each agent has a specific responsibility, and the app can display the trace so a reviewer sees how the final output was assembled.

Video direction

Animate the agents one at a time. Use a distinct icon and one short action word for each: Understand, Retrieve, Analyze, Formulate, Challenge, Explain.

07. The experimental formulation engine

The formulation engine turns the user case and retrieved evidence into a structured experimental proposal. It is more than a block of generated text: it produces named formulations, ingredient amounts, calculations, warnings, and evidence links.

The three formulation roles

Label	Purpose	How to explain it
B0 - Baseline	A reference formulation used as the starting point for comparison.	What happens before the active algae change?
A0 - Control	A controlled comparison that isolates an important variable.	A fair comparison case.
Candidate	The personalized experimental algae-based proposal.	The formulation designed for this user's case.

What the engine should show

- Ingredient list and quantities.
- Target batch size and calculated totals.
- Nutrition-related estimates where supported by data.
- Allergen and restriction checks.
- Evidence links and explanation notes.
- Warnings when information is missing, uncertain, or unsuitable.

Do not oversell the result

The generated formulation is an experimental research proposal. It is not automatically safe for consumption, clinically validated, regulatory approved, or ready for manufacturing.

08. What the user sees in the Flutter app

Screen	What it communicates
1. Start / profile	The user begins a new run and enters personal goals and basic profile information.
2. Preferences and restrictions	The user selects diet rules, disliked ingredients, allergies, and other constraints.
3. Review and submit	The app summarizes the input before sending it to the backend API.
4. Live run progress	The user sees the six-agent trace and the current processing stage.
5. Formulation result	The app displays the baseline, control, candidate recipe, calculations, and warnings.
6. Evidence report	The final report explains why the proposal was made and shows evidence references.
7. History and retry	Previous runs are stored locally. Failed or interrupted runs can be retried safely.
8. Backup and restore	Run history can be backed up and restored for development continuity.

Best screen-recording order

Create a fresh run, enter one realistic sample profile, submit it, pause briefly on each agent trace, open the completed formulation, scroll through the evidence report, then finish on the run-history page.

09. The backend and reliability features

Use this section if the audience wants to know whether the system really works.

The backend exposes an API used by the Flutter app. It validates requests, runs the retrieval and agent workflow, saves the result, and returns status and report data to the user interface.

Reliability already built into the development system

Feature	Why it matters
Input validation	Bad or incomplete requests are rejected early.
SQLite persistence	Runs, stages, and reports remain available after restart.
Run history	Users can return to earlier completed or failed work.
Retry support	Interrupted or failed processing can be attempted again.
Idempotency	The same request is protected from accidental duplicate processing.
Backup and restore	Local run data can be exported and recovered.
Automated tests	Backend and Flutter behavior are checked repeatedly.

Current verification snapshot

- 161 backend tests pass.
- 15 Flutter tests pass.
- Flutter static analysis and development build checks are clean.
- The real local MiniLM-based RAG path has been verified.
- Backup and restore have been verified.
- The production-readiness check intentionally fails until scientific and governance approvals are complete.

10. What is real now, and what still needs work

Ready in the local development build	Still needed before production
End-to-end Flutter-to-API workflow	Scientific expert validation of every production dataset record
Six-agent trace	Formal dataset approval and release governance
Local semantic RAG retrieval	Security, privacy, and threat review
Experimental formulation output	Clinical, food-safety, and regulatory review where applicable
Evidence-linked report	Deployment, monitoring, scaling, and incident procedures
Persistence, retry, history, backup	Production-grade authentication and access control
Automated development verification	Real-world validation with approved users and protocols

The honest conclusion

AlgaMask is a functioning, testable development application with a complete vertical workflow. It is not yet a scientifically approved or production-deployed health product. That distinction builds trust and should be stated clearly in the video.

Recommended final message

AlgaMask demonstrates how evidence retrieval, specialist AI agents, and structured formulation logic can work together to support transparent algae-food research. The next major milestone is governed scientific validation for responsible real-world use.

11. Foolproof 7-10 minute video storyboard

Time	Scene	Editor should show	Voice-over idea
0:00-0:25	Hook	Show algae, food research, and the app title.	Can one system connect personal needs, scientific evidence, and experimental algae-food design?
0:25-0:55	Problem	Scattered papers, generic chatbot response, warning icons.	Personalized formulation is complex, and unsupported AI answers are difficult to trust.
0:55-1:25	Solution	Simple system flow from user to report.	AlgaMask is a research-support app that creates a traceable experimental proposal.
1:25-2:05	User input	Record the questionnaire and review screen.	The user provides goals, diet, allergies, preferences, and constraints.
2:05-2:55	Dataset and RAG	Animate a library, search funnel, and selected evidence cards.	RAG searches the scientific knowledge base before an explanation is generated.
2:55-4:05	Six agents	Reveal each agent and its action word.	Six specialists understand, retrieve, analyze, formulate, challenge, and explain.
4:05-5:05	Formulation	Show B0, A0, candidate, ingredients, and warnings.	The engine creates structured experimental comparisons, not only free text.
5:05-5:55	Report	Scroll citations, trace, cautions, and final summary.	Every important explanation should remain connected to evidence.
5:55-6:35	Reliability	Show history, retry, and backup screens.	Runs are saved, recoverable, and tested across the app and backend.
6:35-7:10	Limits	Use a calm checklist: development ready, approval pending.	Scientific validation and production governance still come before real-world deployment.
7:10-7:35	Close	Return to the full architecture and logo.	AlgaMask is a transparent foundation for evidence-guided algae-food research.

12. Complete presenter script

The editor can shorten pauses or remove sections to match the final runtime.

Opening

AlgaMask is a research-support application for personalized algae-based food formulation. It combines user information, a structured scientific dataset, retrieval-augmented generation, six specialist software agents, and a traceable report.

The problem

Creating a personalized formulation is not as simple as asking a chatbot for a recipe. The system must consider dietary goals, allergies, preferences, ingredient properties, nutrition, formulation constraints, and the strength of the scientific evidence. AlgaMask is designed to make that process structured and visible.

The user journey

The process begins in a Flutter application. The user enters goals, dietary rules, allergies, dislikes, and other constraints. The app reviews this information and sends a structured request to the backend.

The evidence layer

Before creating an explanation, AlgaMask searches its scientific knowledge base using RAG, or retrieval-augmented generation. In simple terms, the system opens the right pages of its scientific library before it starts reasoning. The current development dataset is organized into 21 tables and 286 records, with traceability back to source information.

The six agents

The Profile Agent understands the person. The Retrieval Agent finds evidence. The Chemistry Agent examines scientific and ingredient considerations. The Recipe Agent builds the experimental formulation. The Critic Agent challenges weak or unsafe parts. Finally, the Impact Agent explains the result, trade-offs, evidence, and cautions.

The formulation

The system produces structured experimental comparisons. B0 acts as a baseline, A0 acts as a control, and the candidate represents the personalized algae-based proposal. The output can include ingredient quantities, calculations, estimated nutrition, allergen checks, evidence links, and warnings.

The report and history

The result is presented as a readable, evidence-linked report. The app also stores run history, supports safe retry, and includes backup and restore features. This means the development workflow can be demonstrated from beginning to end.

The honest limitation

AlgaMask is currently a functioning local development system. It is not a doctor, a medical diagnosis tool, or an approved commercial food product. Scientific experts still need to validate the production dataset, and security, regulatory, deployment, and governance work must be completed before real-world production use.

Closing

The value of AlgaMask is transparency. It shows how personal requirements, scientific evidence, specialist AI reasoning, and formulation logic can work together in one inspectable workflow. It is a strong foundation for responsible, evidence-guided algae-food research.

13. Editor checklist and terminology

Before editing

- Capture a successful app run from the questionnaire to the final report.
- Keep one sample user profile consistent throughout the video.
- Prepare six simple icons for the agents and one icon for the RAG library.
- Use readable labels and avoid dense source code on screen.
- Blur personal or sensitive information in recordings.

During editing

- Show the full flow before explaining individual components.
- Use the same colors for RAG, agents, formulation, and report in every scene.
- Keep the agent trace visible long enough to prove the workflow is real.
- Add a clear on-screen disclaimer when the formulation result appears.
- End with the current achievement and the next scientific-validation milestone.

Words to use and avoid

Prefer	Avoid
Research-support application	AI doctor
Experimental formulation	Proven treatment
Evidence-linked explanation	Guaranteed truth
Development dataset	Fully approved medical database
Potential purpose or estimated value	Guaranteed health benefit
Scientific approval pending	Ready for public production

Final quality check

A viewer with no coding knowledge should be able to answer five questions after watching: What problem does AlgaMask solve? What information does the user provide? Why is RAG used? What do the six agents do? What is complete today, and what still requires approval?